

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID - SIH26038

Problem Statement Title - Explainable AI for Diabetic Retinopathy Screening in Rural India

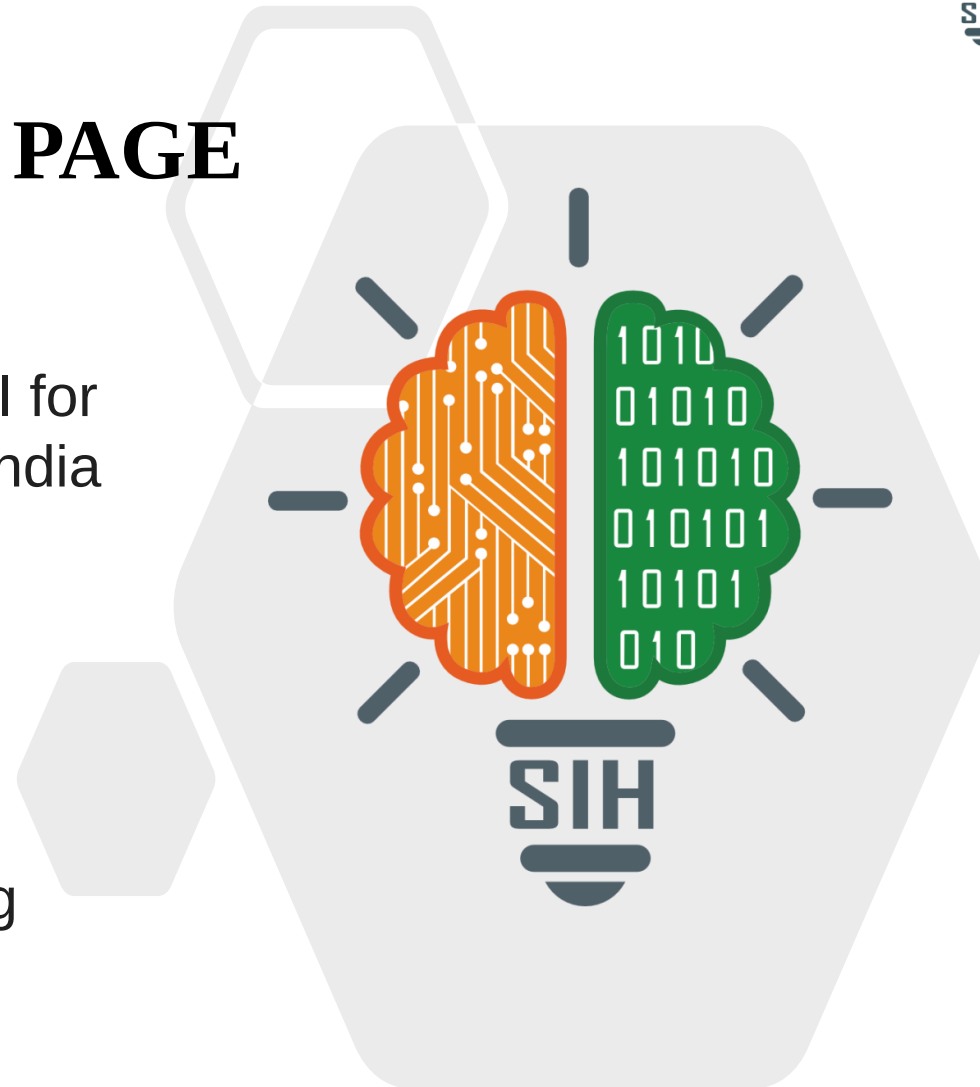
Theme - MedTech / BioTech / HealthTech

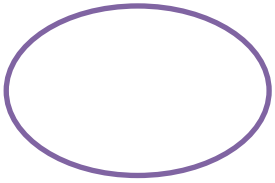
PS Category - Software

Team ID - 148873

Team Name - MMC

Institute - Motihari College of Engineering



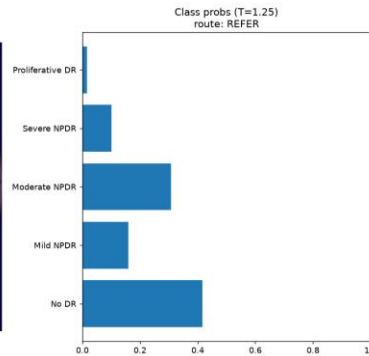
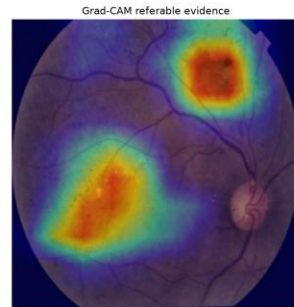


IDEA TITLE

An explainable, quality-aware AI workflow that grades diabetic retinopathy, surfaces retinal evidence, generates an AI-assisted screening report, and routes each case to screening output, human review, or referral.



Case 2: referable_good_quality | true Proliferative DR | route REFER



Real explainability hero case - referable DR (Grad-CAM + AI report snippet)

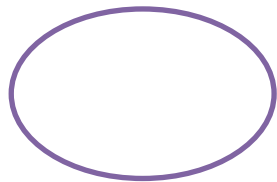
(1) Quality-aware screening

(2) Evidence-aware explainability

(3) MATLAB + Simulink workflow integration

AI-ASSISTED SCREENING $\neq$ AUTONOMOUS DIAGNOSIS

Clinician oversight + further clinical validation remain required.



TECHNICAL APPROACH

METRICS - ONE-SHOT LOCKED TEST · APTOS 2019 · n = 732

0.9027

Quadratic Weighted Kappa

91.92%

Referable Sensitivity

93.33%

Specificity

0.9792

ROC AUC

IMAGE QUALITY

Heuristic quality index
gate → accept / review

RETINAL STRUCTURES

OD Dice 0.897 - STRUCTURAL EVIDENCE
fovea 273 px · vessels - PROTOTYPE

LESION EVIDENCE

MA · HE · EX · SE
mean Dice 0.409 - PROTOTYPE

DR GRADING

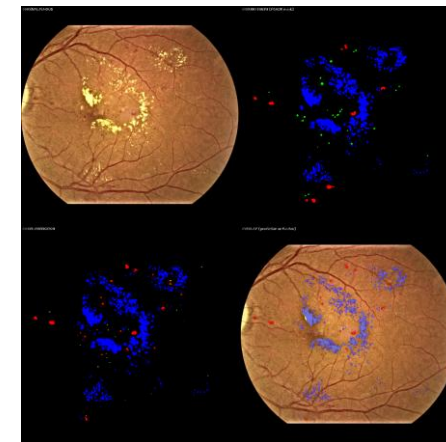
ICDR 0-4 · QWK 0.9027
T=1.25 temp. scaling + argmax

EXPLAINABILITY

Grad-CAM attention evidence
AI-assisted screening report

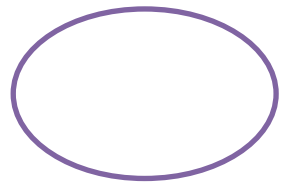
MATLAB / SIMULINK

ONNX export · parity 6.7e-6
queue / workflow simulation



FGADR prototype - EX case 1091_2
Dice 0.871 (PROTOTYPE)

MATLAB / ONNX parity max $\approx 6.7e-6$ · Python → ONNX $\approx 8.1e-6$ · MATLAB reproduced all 4 routing cases
Lesion / fovea / vessel components are labelled PROTOTYPE. Optic-disc Dice 0.897 is STRUCTURAL EVIDENCE, not lesion evidence.



FEASIBILITY AND VIABILITY



VALIDATED NOW

- **Locked APTOS test n=732: QWK 0.9027 · Sens 91.92% · Spec 93.33%**
- MATLAB / ONNX parity $\approx 6.7e-6$ · 4/4 routing cases reproduced
- 0 missed referable cases (evaluated workflow)
- Simulink LOW / BASE / HIGH queue scenarios drained

PROTOTYPE EVIDENCE

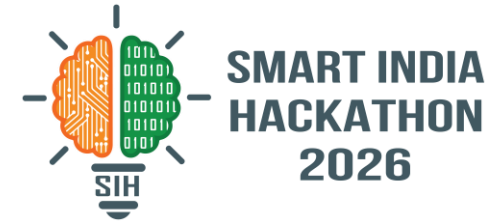
- **Optic-disc structural Dice 0.897**
- Fovea prototype: median 273 px · 32% within 200 px
- FGADR lesion evidence: mean Dice 0.409
- Real AI-assisted reports · Grad-CAM + lesion overlays

REMAINING VALIDATION

- **Clinician reviewer study (pending)**
- Prospective clinical validation (pending)
- NV - insufficient data (experimental-only)
- Broader external-domain validation
- Field deployment / offline hardware validation

NEXT VALIDATION: blinded clinician review of AI report + routing + evidence overlays (pending - not yet executed)

IMPACT AND BENEFITS



PHC / EYE CAMP

LOCAL AI SCREENING
(AI-assisted prototype)

AI EVIDENCE + REPORT

CLINICIAN REVIEW

DISTRICT REFERRAL

91.92%
referable sensitivity

93.33%
specificity

161
human-review cases

302
referral cases

≈ 346,000 screenings / year - MODELED CAPACITY
Simulink queue/workflow simulation only - not deployed or clinically validated capacity.

Technician

Instant quality check + evidence pack
at PHC / camps

Clinician

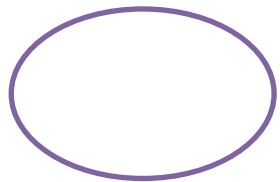
AI-assisted report with explainable
overlays, remote review

District system

Standardized routing + referral queue,
reduced specialist load

HUMAN-IN-THE-LOOP: AI screens and routes; clinician reviews; referral is clinician-driven.

AI-assisted screening report ≠ autonomous diagnosis



RESEARCH AND REFERENCES



A. RESEARCH / EVIDENCE

APTOS 2019 -

Kaggle blindness challenge - locked test n = 732

FGADR -

arXiv:2008.09772 - lesion prototype (non-commercial use)

IDRiD -

IEEE JBHI 2020 - external OD / fovea + lesion reference

DRIVE -

vessel prototype (dev split only; not official test)

ICDR 5-point scale -

Wilkinson et al., Diabetes Care 2003

Grad-CAM -

Selvaraju et al., CVPR - classifier attention evidence

ONNX → MATLAB -

parity $\approx 6.7e-6$ - PyTorch model exported via ONNX

Simulink -

queue / workflow simulation (modeled capacity)

B. LIMITATIONS / NEXT STEP

- Clinician reviewer validation - pending
- NV evidence - insufficient data (experimental-only)
- Lesion segmentation - research prototype (MA 0.256 weakest)
- External / domain-shift risk - broader validation required
- Prospective clinical validation - required before deployment